

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	2025	Date:	

Code of Conduct

I **YUVARAJ R / 192524058** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Yuvaraj R

Instructions

- Read each industry scenario carefully before answering.
- Analyze the given problem from a semantic analysis perspective.
- Provide structured and logical answers with proper justification.
- Support your analysis using examples from the given scenario wherever applicable.
- Use suitable diagrams, semantic representations, logical predicates, workflows, architectures, or tables whenever necessary.
- Clearly state assumptions if additional information is required.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic interpretation of user queries, identify ambiguity in meaning representation, evaluate semantic processing limitations, and improve understanding in banking chatbot applications.	Q1
Syntax-Driven Semantic Analysis	Analyze multiple interpretations of spoken commands, evaluate parsing-related ambiguity, and assess semantic extraction in real-time voice assistant systems.	Q2
First-Order Predicate Calculus, Representing Linguistically Relevant Concepts, Syntax-Driven Semantic Analysis	Design a semantic processing architecture for healthcare reports, model relationships among entities and actions, represent linguistic concepts, and improve semantic extraction accuracy.	Q3

Annexure A – Industry Problem Solving Task

1. AI-Powered Insurance Claim Processing System

An insurance company is developing an NLP-based system to automatically process customer insurance claims. The system uses **Context-Free Grammar (CFG)** to parse customer statements and extract information such as the claimant, damaged object, cause of damage, and claim amount.

However, the system faces the following problems:

- Ambiguous sentence structures
- Difficulty distinguishing noun and prepositional phrase attachment
- Subject–verb agreement errors
- Inefficient parsing of lengthy customer descriptions
- Difficulty handling conversational and incomplete input

Example customer statement:

"I reported the damage to the car after the accident."

The sentence can produce different interpretations depending on whether "**to the car**" is associated with *reported* or *damage*.

Tasks:

- a) Analyze the structural ambiguity in the given sentence using CFG and illustrate the possible interpretations.
- b) Evaluate the limitations of a basic CFG in handling ambiguity, agreement, and long customer-generated insurance statements.
- c) Design an improved parsing approach using **PCFG, Feature Structures, and Earley Parsing** to resolve ambiguity and efficiently process the claim.
- d) Justify how the proposed architecture can improve the accuracy, scalability, and reliability of automated insurance claim processing.

2. Intelligent Railway Ticket Booking Assistant

A railway company is developing a conversational AI assistant for ticket booking. The system currently uses **top-down parsing** to interpret passenger requests.

The system encounters:

- Excessive backtracking
- Ambiguous attachment of phrases
- Incomplete passenger requests
- Long-distance dependencies
- Slow response for complex queries

Example passenger request:

"Book two tickets from Chennai to Delhi for my parents with AC sleeper."

The system sometimes incorrectly associates "**with AC sleeper**" with the passengers instead of the ticket/class information.

Tasks:

- a) Analyze the possible syntactic interpretations of the given passenger request using CFG.
- b) Evaluate why top-down parsing may result in excessive backtracking and inefficient processing for conversational ticket-booking queries.
- c) Analyze how **Earley Parsing** can handle ambiguity, incomplete input, and different possible parse structures more effectively.
- d) Compare **Top-Down Parsing, CFG-based parsing, and Earley Parsing** in terms of ambiguity handling, computational efficiency, partial-input processing, and suitability for real-time railway booking systems.

3. AI-Based Legal Contract Analysis System

A legal technology company is developing an NLP system to analyze contracts and automatically identify:

- Parties involved

- Obligations
- Conditions
- Deadlines
- Actions
- Exceptions

The existing system uses a basic CFG parser. However, it produces incorrect interpretations when processing long legal sentences containing relative clauses, passive constructions, and multiple prepositional phrases.

Example contract sentence:

"The supplier who delivered the equipment to the company must replace the defective components within thirty days."

The system sometimes incorrectly identifies the **company** as the entity responsible for delivering the equipment.

Tasks:

- Analyze the syntactic structure of the given sentence using CFG and identify possible sources of ambiguity.
- Evaluate why basic CFG is insufficient for accurately interpreting complex legal sentences containing relative clauses and nested structures.
- Design an NLP architecture integrating **CFG, PCFG, Feature Structures, and Earley Parsing** to improve syntactic and semantic interpretation.
- Develop a step-by-step processing workflow showing how the system can extract:
 - Supplier
 - Action
 - Equipment
 - Company
 - Obligation
 - Deadline

and justify how the proposed approach improves automated contract analysis.

4. AI-Based E-Commerce Product Search System

An e-commerce company is developing a conversational product-search system. Customers enter natural-language queries rather than selecting predefined filters.

The current system uses CFG but faces difficulties with:

- Ambiguous phrase attachment
- Coordination
- Missing words
- Informal customer queries
- Long product descriptions
- Real-time search requirements

Example user query:

"Find laptops with a touchscreen for students under ₹60,000."

The system may incorrectly associate "**for students**" with the touchscreen rather than the laptop.

Tasks:

- Analyze the syntactic ambiguity in the given query and construct possible parse interpretations using CFG.
- Evaluate the limitations of CFG in handling informal, elliptical, and ambiguous e-commerce search queries.
- Propose an improved parsing architecture using **PCFG, Feature Structures, and Earley Parsing** to identify product attributes, user requirements, and price constraints.
- Justify how the proposed solution can improve search precision, response time, and customer experience in a large-scale e-commerce platform.

5. Intelligent Airline Voice Assistant

An airline is developing a real-time voice assistant that allows passengers to modify flight bookings using spoken commands.

The system currently uses a conventional top-down parser. It experiences:

- Backtracking
- Ambiguous attachment
- Speech recognition errors
- Incomplete commands
- Delayed responses
- Difficulty interpreting corrections made by users

Example spoken command:

"Change my flight to Mumbai tomorrow with two passengers."

Depending on the parse, "**with two passengers**" may be incorrectly interpreted as an attribute of the destination rather than the number of passengers.

The passenger may also say:

"Change my flight to Mumbai tomorrow... actually, make that Bangalore."

Tasks:

- a) Analyze the possible syntactic structures and ambiguities in the given voice command.
- b) Evaluate the limitations of top-down parsing when processing incomplete, corrected, and ambiguous spoken commands in real time.
- c) Design a parsing architecture using **Earley Parsing, PCFG, and Feature Structures** to support partial input, ambiguity resolution, and agreement constraints.
- d) Develop a processing workflow showing how speech input can be converted into a structured booking request and justify the proposed method for real-time airline applications.

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools

Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Q. No. / Criteria Level	Understanding of Industry Problem	Application of Engineering Concepts	Solution Design	Use of Tools	Analysis	Professionalism	Sub. Total	Total %
1								
2								
3								
4								
5								

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. AI-Powered Insurance Claim Processing System

Given sentence: “I reported the damage to the car after the accident.”

a) Structural ambiguity using CFG

The sentence has prepositional phrase (PP) attachment ambiguity. The phrase “to the car” can be associated with “damage” or with “reported”.

$S \rightarrow NP VP$

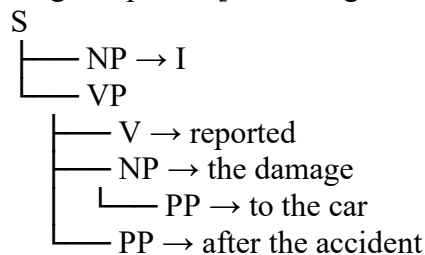
$VP \rightarrow V NP PP$

$NP \rightarrow Det N$

$PP \rightarrow P NP$

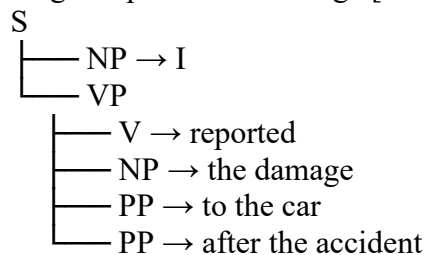
Interpretation 1 – “to the car” modifies the damage:

Meaning: I reported [the damage to the car] [after the accident].



Interpretation 2 – “to the car” is attached to the reporting action:

Meaning: I reported the damage [to the car] [after the accident].



b) Limitations of basic CFG

- CFG can generate multiple valid parse trees without deciding which interpretation is correct.
- Basic CFG does not naturally enforce subject–verb agreement constraints.
- Long customer descriptions can produce many possible parse structures.
- Conversational and incomplete inputs are difficult to process using only basic CFG.
- Exploring many possible parses can increase processing time.
- CFG mainly represents syntax and does not by itself use contextual meaning to choose the intended interpretation.

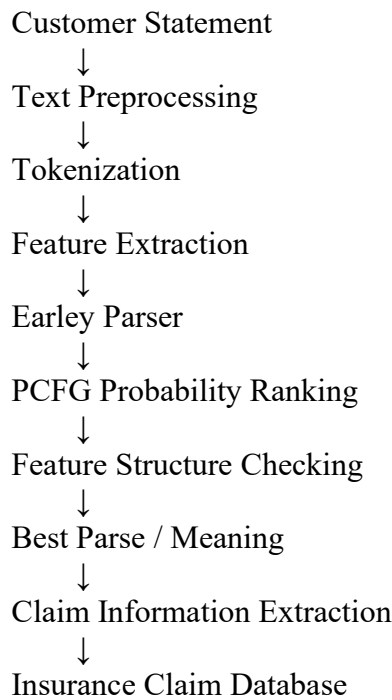
c) Improved parsing approach

A robust solution combines PCFG, Feature Structures, and Earley Parsing.

PCFG assigns probabilities to grammar rules so that alternative interpretations can be ranked.

Feature Structures represent grammatical information such as number, person, and tense and help enforce agreement constraints.

Earley Parsing maintains and evaluates multiple possible structures and is suitable for ambiguous or incomplete input.



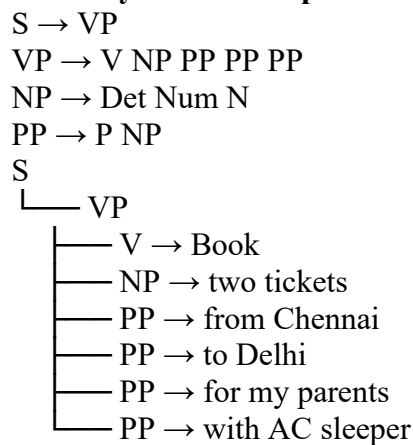
d) Justification

- Accuracy: PCFG ranks alternative interpretations and Feature Structures enforce linguistic constraints.
- Scalability: Earley Parsing can handle varied and incomplete customer statements.
- Reliability: Combining syntactic and semantic information reduces incorrect extraction of claimant, damaged object, cause, and claim details.
- The architecture is therefore more suitable for automated insurance claim processing than a basic CFG-only system.

2.Intelligent Railway Ticket Booking Assistant

Given request: “Book two tickets from Chennai to Delhi for my parents with AC sleeper.”

a) Possible syntactic interpretations using CFG



The intended interpretation is that “with AC sleeper” describes the ticket/class. A possible incorrect parse may associate “with AC sleeper” with the passengers. This is an attachment ambiguity.

b) Why Top-Down Parsing may cause excessive backtracking

- Top-down parsing begins from the start symbol and predicts possible sentence structures.
- If a prediction fails, the parser may return and try another grammar rule.

- Ambiguous phrases create several possible structures and increase backtracking.
- Long-distance dependencies make prediction more difficult.
- Incomplete conversational requests may require additional alternatives.
- For complex booking queries, this can increase processing time and reduce real-time responsiveness.

c) Earley Parsing for ambiguity and incomplete input

Earley Parsing is suitable because it can maintain multiple possible parse states while processing the input. It can continue with incomplete input and later use additional words to determine the most appropriate structure.

Input
 ↓
 Book two tickets
 ↓
 from Chennai
 ↓
 to Delhi
 ↓
 for my parents
 ↓
 with AC sleeper
 ↓
 Maintain / evaluate possible parse structures

- Handles ambiguous grammar.
- Supports incomplete input.
- Can handle different sentence structures.
- Reduces unnecessary repeated parsing through chart-based processing.
- Is suitable for conversational NLP systems.

d) Comparison

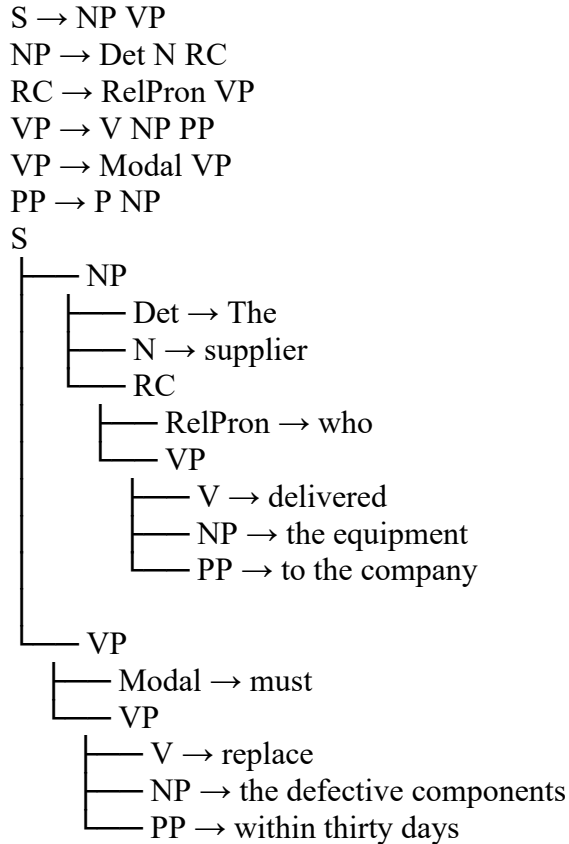
Feature	Top-Down Parsing	CFG-Based Parsing	Earley Parsing
Ambiguity	Difficult	Can generate multiple parses	Handles multiple parses
Backtracking	Can be high	Depends on parser	Chart-based processing reduces repeated work
Incomplete input	Limited	Limited	Better suited
Complex queries	Can be slow	May produce many parses	More suitable
Real-time use	Less suitable	Moderate	More suitable

Conclusion: Earley Parsing is more suitable for a conversational railway booking assistant because it handles ambiguity and incomplete requests more effectively.

3. AI-Based Legal Contract Analysis System

Given sentence: “The supplier who delivered the equipment to the company must replace the defective components within thirty days.”

a) Syntactic structure using CFG

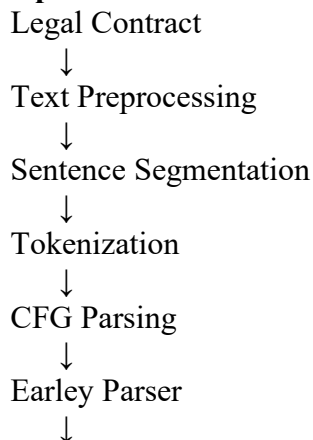


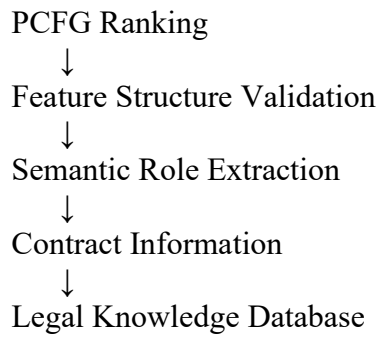
A possible source of ambiguity is the attachment of the prepositional phrase “to the company”. Incorrect attachment can cause the system to assign the wrong entity as the responsible party.

b) Why basic CFG is insufficient

- Legal sentences often contain relative clauses and nested structures.
- Multiple prepositional phrases can create attachment ambiguity.
- Basic CFG does not directly represent semantic roles such as who performed an action.
- Long legal sentences create many possible parse structures.
- Accurate identification of parties, obligations, conditions, actions, exceptions, and deadlines requires more than syntax alone.

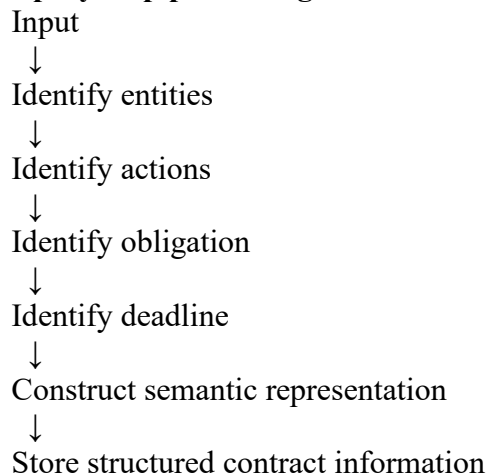
c) Proposed NLP architecture





- CFG provides the basic grammatical structure.
- Earley Parsing handles long, nested, and ambiguous structures.
- PCFG ranks alternative interpretations.
- Feature Structures maintain grammatical and linguistic constraints.
- Semantic extraction identifies entities, actions, obligations, and deadlines.

d) Step-by-step processing workflow



Extracted information:

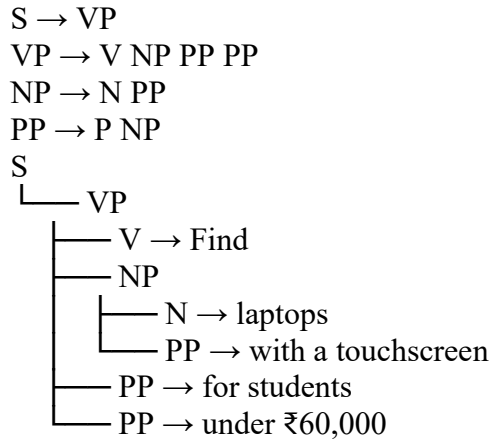
Information	Extracted value
Supplier	The supplier
Action	Replace
Equipment	The equipment
Company	The company
Obligation	Must replace defective components
Deadline	Within thirty days

The proposed approach improves automated contract analysis by reducing incorrect entity-action relationships and improving interpretation of complex legal sentences.

4. AI-Based E-Commerce Product Search System

Given query: “Find laptops with a touchscreen for students under ₹60,000.”

a) Syntactic ambiguity using CFG



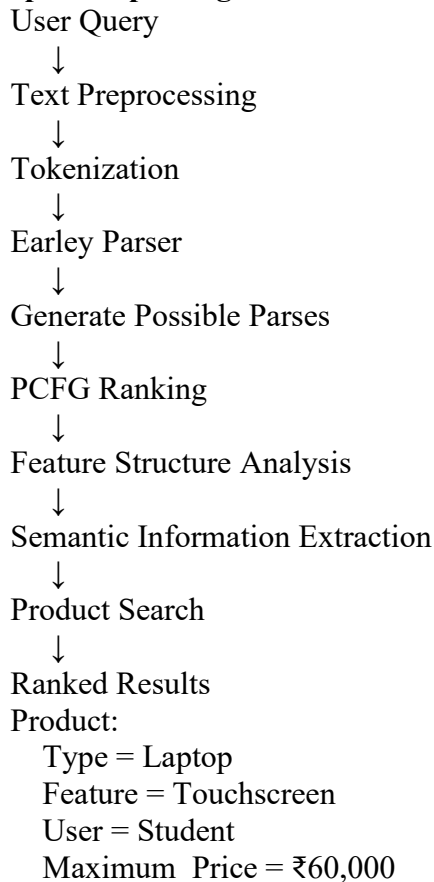
Correct interpretation: Product = laptop, Feature = touchscreen, Target user = students, Maximum price = ₹60,000.

Possible incorrect interpretation: “for students” could be attached to “touchscreen” instead of to the laptop.

b) Limitations of CFG

- Ambiguous phrase attachment.
- Difficulty with coordination.
- Difficulty with missing words and elliptical queries.
- Informal customer queries may not follow complete grammar rules.
- Long product descriptions may produce many possible structures.
- Real-time product search requires efficient interpretation.

c) Improved parsing architecture



d) Justification

- Search precision improves because product type, feature, user requirement, and price are separated correctly.
- PCFG helps rank likely interpretations.
- Earley Parsing supports different and incomplete query structures.
- Structured constraints can be passed to the product-search engine.
- Customers can search naturally instead of selecting many predefined filters.
- The approach improves search precision, response time, and customer experience.

Product = Laptop

Touchscreen = Yes

Target = Students

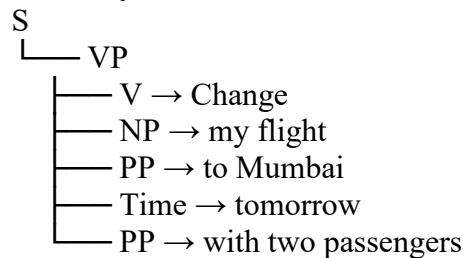
Price $\leq$ ₹60,000

5. Intelligent Airline Voice Assistant

Given command: “Change my flight to Mumbai tomorrow with two passengers.”

Correction example: “Change my flight to Mumbai tomorrow... actually, make that Bangalore.”

a) Possible syntactic structures and ambiguities



Correct interpretation: Action = Change; Destination = Mumbai; Date = Tomorrow; Passengers = 2.

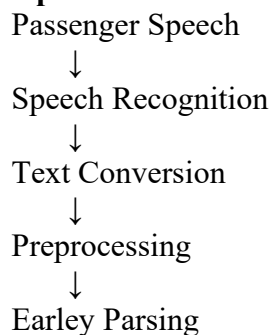
A possible incorrect interpretation may treat “with two passengers” as an attribute of the destination instead of as the passenger count.

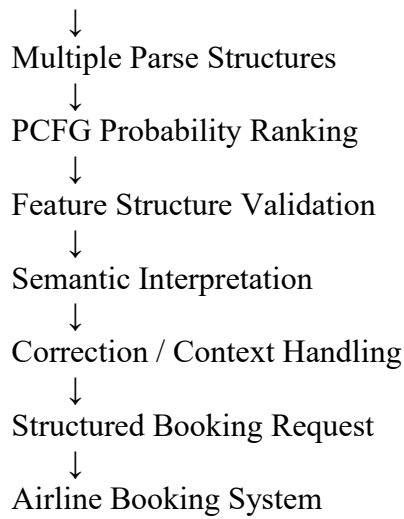
For the correction, the final destination must be updated from Mumbai to Bangalore.

b) Limitations of Top-Down Parsing

- Backtracking can increase for ambiguous spoken commands.
- Speech recognition errors can produce unexpected input.
- Incomplete commands are difficult to interpret.
- User corrections require the system to revise an earlier interpretation.
- Real-time voice applications require fast responses.
- Ambiguous attachment can cause incorrect booking information.

c) Proposed architecture



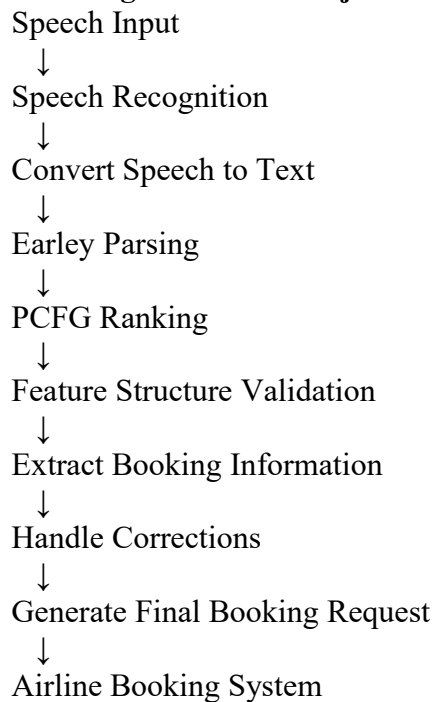


Action = Change Flight
Destination = Mumbai
Date = Tomorrow
Passengers = 2

After the correction “actually, make that Bangalore”, the destination feature is updated:

Destination:
Mumbai → Bangalore

d) Processing workflow and justification



Final structured request:
Action: Change Flight
Destination: Bangalore
Date: Tomorrow
Passengers: 2

- Supports partial and incomplete input.
- Handles alternative parse structures.
- PCFG helps resolve ambiguity by ranking interpretations.
- Feature Structures support grammatical and agreement constraints.
- Correction handling allows later speech to replace an earlier value.
- The combined architecture is suitable for real-time airline applications.

